

Logan Burns

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Education

University of Florida, B.S. Electrical Engineering

Herbert Wertheim College of Engineering, GPA 3.92, University Honors Program

Expected May 2028

Gainesville, FL

- Relevant coursework: Digital Design, Microprocessor Applications, Electronic Circuits, Signals & Systems, Circuits I/II.

Experience

Head Peer Instructor, EEL 3701C Digital Logic & EEL 4744C μ P Applications

Leads the peer-instructor team for both courses, UF ECE

Spr 2026 – Present

Gainesville, FL

- Manage exam-grading assignments across the PI team, course hardware inventory, and hardware-repair coordination; point of contact for fellow peer instructors.
- Lead help sessions on external-bus memory mapping, address decoding, and digital logic design; author lab specifications, solutions, and course documentation.
- Resolved a Quartus–Questa incompatibility with a C++ command-line shim distributed to students to restore local HDL simulation; built a JTAG batch-programming and QC pipeline for the course's DE10-Lite FPGA fleet.
- Remediated 150 course documents for ADA/WCAG accessibility with a custom Python toolchain (contrast analysis, document/PDF tagging, GUI, packaged executable); automated Canvas grade entry with a verified-diff Selenium tool.

Florida Optics and Computational Sensor (FOCUS) Lab, UF ECE

Undergraduate Research Assistant, embedded systems & thermal instrumentation

Sep 2025 – Present

Gainesville, FL

- Built a calibrated FLIR A6751sc thermal-imaging pipeline (GigE Vision) and a 150 W radiant-heat stress rig to characterize ML workloads on a Jetson Orin NX across DVFS operating points with per-rail power telemetry; work contributed to a co-authored manuscript in revision for conference resubmission, including validation of a controller achieving 16.3°C lower peak temperature at equal accuracy.
- Sole author of a virtual thermal sensing system predicting temperatures of off-die board components (VRMs, MOSFETs) from on-die telemetry: thermal response mapping, lumped-capacitance model fitting, and a real-time Linux daemon (under 1% CPU) needing no camera after a one-time calibration. Predicts 16 board hotspots to 0.34°C mean error on held-out operating points.
- Extended it from point prediction to the full thermal field: eight POD modes capture 99.8% of the spatial variance, reconstructing every pixel to 0.41°C mean error on held-out configurations.

Machine Intelligence Laboratory (MIL), UF: SubjuGator 9 AUV

Electrical Team, sole owner of the leak-detection subsystem

Jan 2026 – May 2026

Gainesville, FL

- Designed the vehicle's water-ingress detection board end to end: LTspice comparator front-end simulation, Altium schematic capture and 2-layer layout, and a fabrication-released design (v2) with full manufacturing outputs (gerbers, NC drill, BOM, pick-and-place).

Projects

Polyphonic Wavetable Synthesizer, bare-metal AVR (C, assembly)

- Real-time 3-voice synthesizer on an XMEGA (20 kHz, 12-bit DAC): gapless audio via ping-pong DMA, inline-assembly fixed-point mixing, voice-stealing allocation, per-voice ADSR/filtering, and a live serial oscilloscope display at 2 Mbaud.

Discrete LDO Voltage Regulator, LTspice (individual design)

- Rebuilt a discrete LDO whose original compensation never let the loop cross unity: a series damping zero below crossover holds phase margin within 0.1° across a 60:1 output-capacitor ESR range, and a P-channel pass device restores regulation at the low input corner (12.6 mV line regulation, 8–16 V). Foldback limiting cuts pass-device dissipation 9.3 W to 2 W.
- Designed a discrete two-stage BJT op-amp: 110 dB open-loop gain, 3.63 MHz GBW, 106° phase margin by nulling the right-half-plane zero emitter degeneration drags toward crossover. Both designs are now under bench characterization.

FPGA Digital Design & Verification, VHDL/SystemVerilog (DE10-Lite)

- Built a complete 2-player Pong entirely in hardware (11-module VHDL design with reusable VGA subsystem); verified designs with constrained-random testbenches, functional coverage, and SystemVerilog Assertions in QuestaSim.

Fabrication Experience

Semiconductor Readiness Organization, UF Nanoscale Research Facility

Sep 2024 – Present

- Fabricated MOS capacitors in the NRF cleanroom: dry oxidation, spin-coating, photolithography, HF wet etch, sputter deposition.

Skills

Languages: C, C++, Python, MATLAB, VHDL, SystemVerilog, AVR assembly

EDA & Simulation: Altium Designer, LTspice, Cadence Virtuoso, Quartus, QuestaSim (SVA, constrained-random verif.)

Embedded & Lab: NVIDIA Jetson, PyTorch, NumPy/pandas, FLIR/GigE Vision, DMA- and interrupt-driven firmware, Linux, Git; thermal imaging, I–V characterization, oscilloscope/AWG/network analysis; SMT reflow assembly; cleanroom microfabrication